

18/8/25

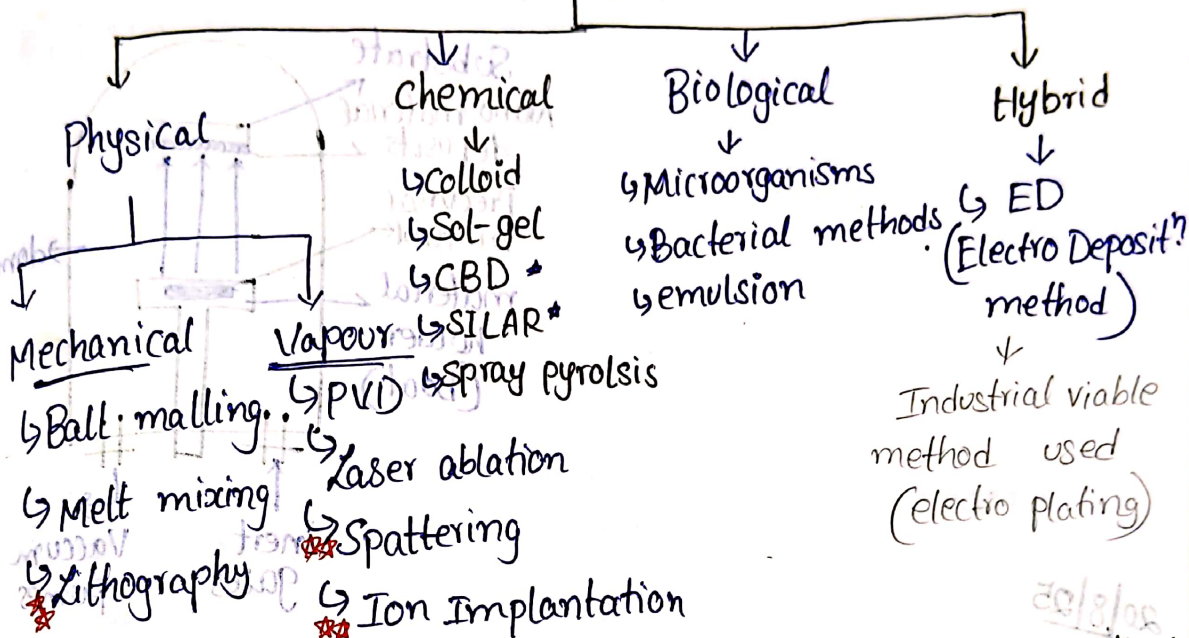
Synthesis Methods

for preparing nanomaterials

Top - Bottom Approach

Bottom - up Approach

~~Lithography~~



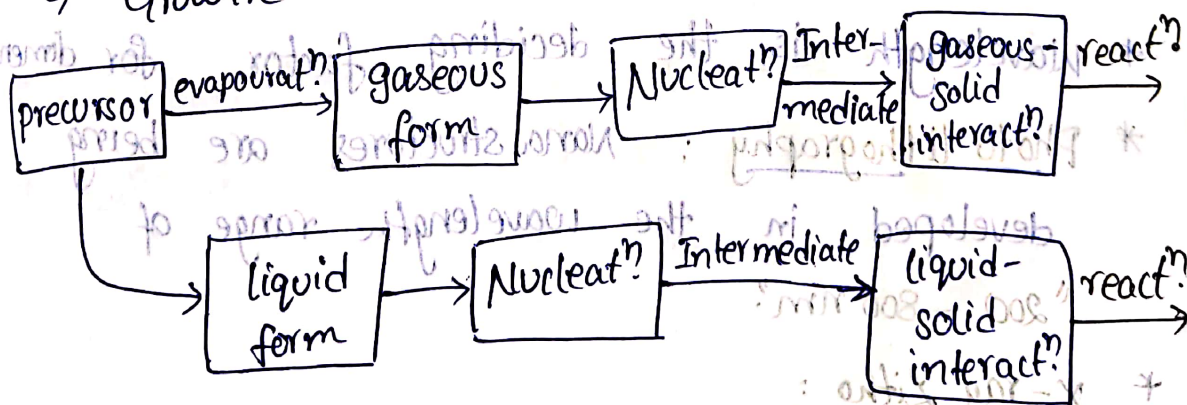
Principles / mechanisms used in Synthesis methods

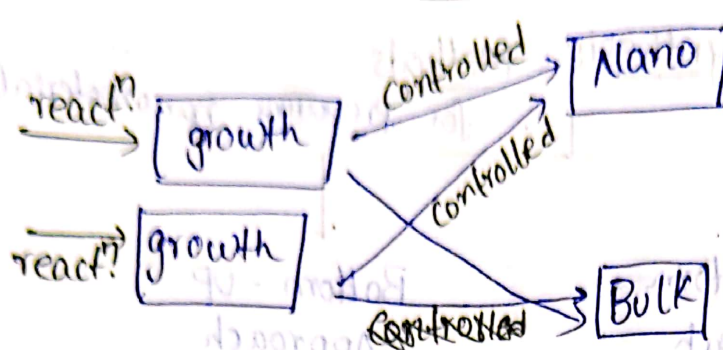
for building nano structures:

↳ Precursor, evaporation / liquid / gaseous

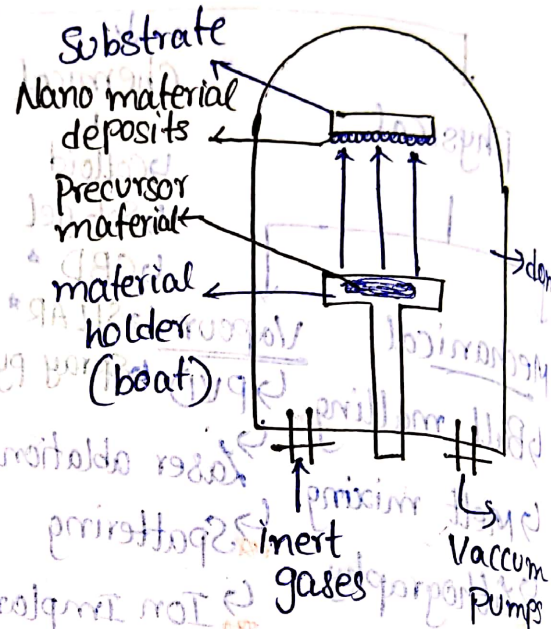
↳ Nucleation

↳ Growth





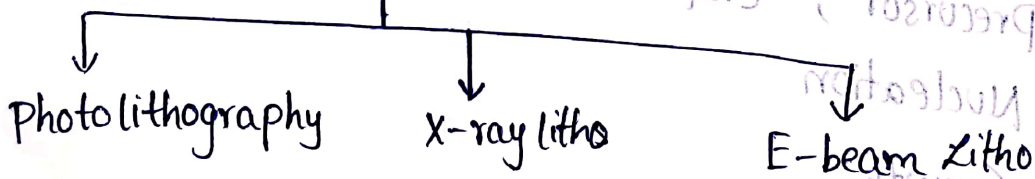
Thermal evaporation method :



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Lithography Process : → for devising Nanostructure

↳ Transferring of geometrical patterns to develop nano structures.



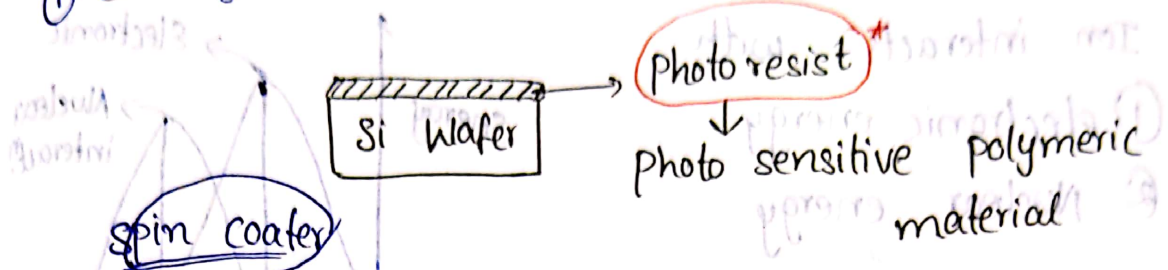
* Wavelength is the deciding factor for dimension

* Photolithography : Nano structures are being developed in the wavelength range of "200 - 800 nm".

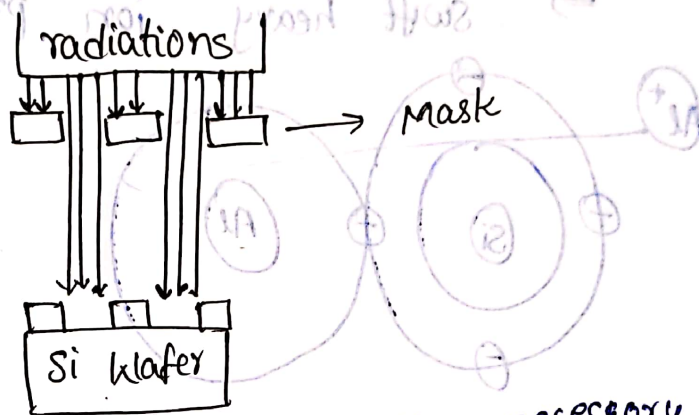
* X-ray Litho :

Steps:

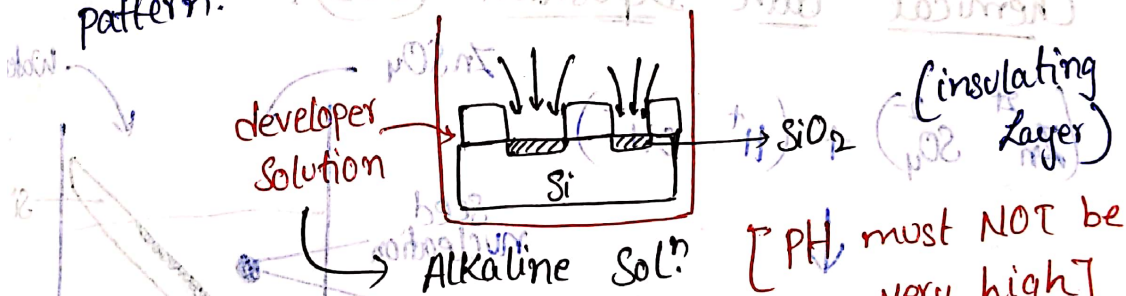
① Coating the photoresist on silicon wafer.



② Exposure to radiations.

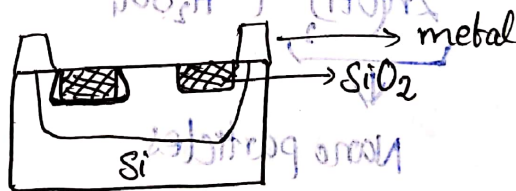


③ Use developer for developing the necessary pattern.



④ Etching Process

HF



* Microfabrication of semi conductors → MEMS

* Nanofabrication → NEMS

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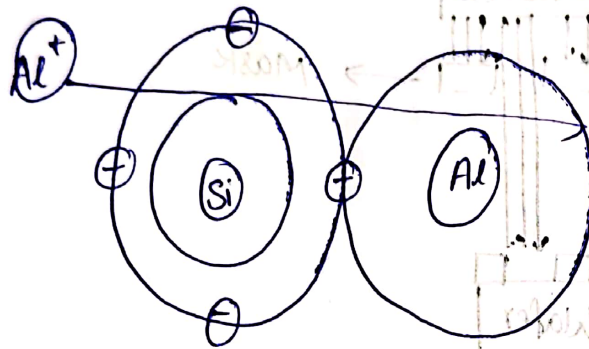
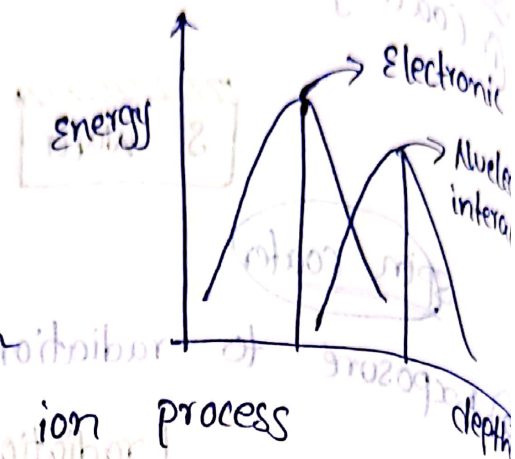
* Ion Implantation

Ion interacts with

- ① electronic energy
- ② Nuclear energy

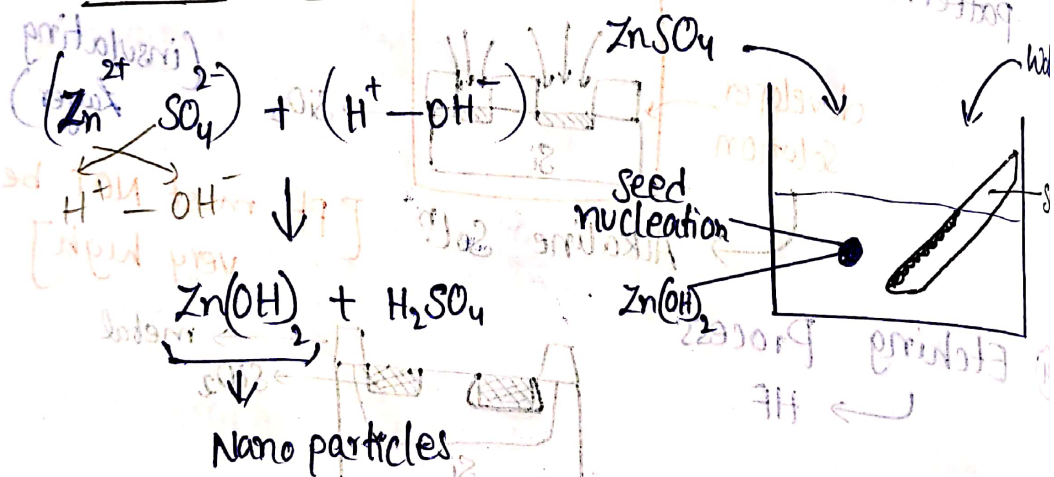
* swift energy is high

→ swift heavy ion process



* CBD is the easiest method used for dopin

* Chemical Bath Deposition (CBD):



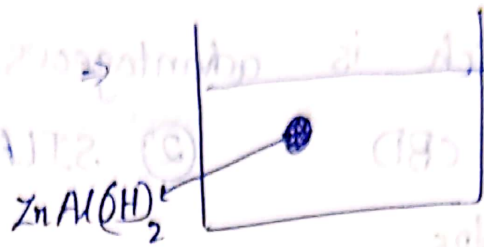
For controlling the growth of $Zn(OH)_2$

- ① Volumetric control of chemicals
- ② Temperature control

① 1:2

① $ZnSO_4$

② Al_2SO_4



↳ Doped with Al

* Crude method for doping is CBD. But it is not precise.

* Ion implantation is precise process.

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Ion implantation

fluence = $\frac{\text{no. of ions}}{\text{cm}^2}$

Ostwald's ripening

CBD

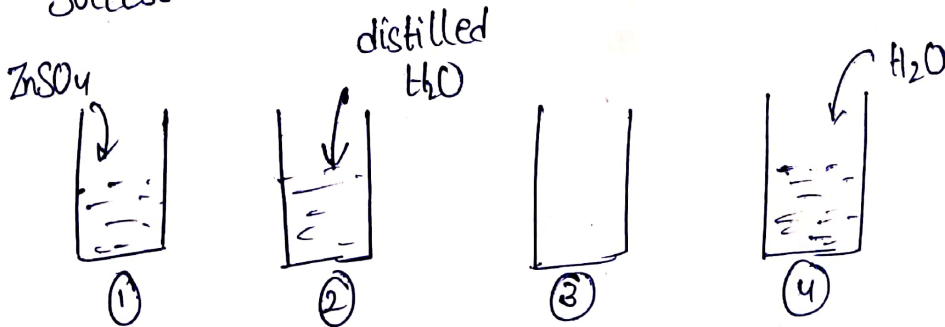
$ZnCl_2$

$ZnSO_4$

$CdSO_4$

SILAR:

Successive Ionic layer adsorption of reaction



Which is advantageous among

- ① CBD ② SILAR ③ Spray

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* Plasmonic state

* Surface Plasmonic Resonance Effect

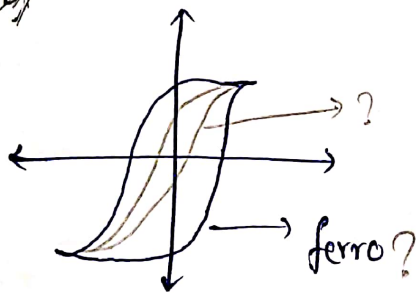
* Magnetic effects

Ferro dia para antiferro ferrimagnetic

Nano effects

super para magnetic? → hysteresis?

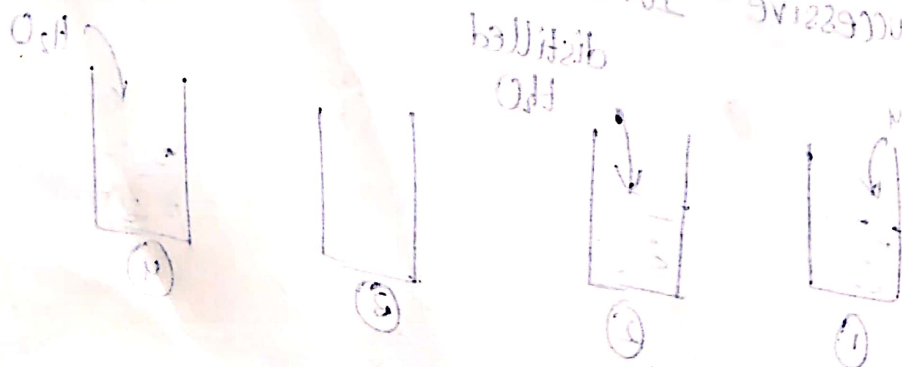
* ~~Magnetic~~ hysteresis



* XRD → structural

XPS → chemical composition

↳ x-ray photoemission spectroscopy



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Photonics

Principle of Atomic force microscopy

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Characterization techniques:

① X-ray diffraction

② Electron microscopy: scanning electron microscopy
Transmission " "

③ Atomic force microscopy

↳ Magnetic force microscopy

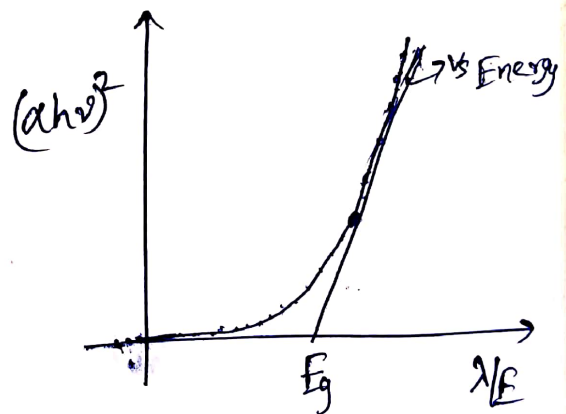
↳ Nano Indentation

* Conductive image

* Spectrophotometer that works in the visible & IR.

UV-Vis spectroscopy

* particle must be either transmitting or absorbing in nature.



$$* \text{Transmission (T)} = \frac{I_t}{I_0} = \frac{1}{\exp(\alpha t)}$$

$$I_0 = I_t \cdot \exp(\alpha t)$$

* Band Gap (in bulk vs nano particles)
(Optical Analysis)

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C = 6

Si = 14

Ge = 32

Sn = 50

Pb = 82

+18 → 2n² (n=3)

+32 (n=4)

* "Si" is mostly used in semiconductors because

i. Availability . 2nd largest element

ii. Temperature stability:

Si 150 - 200 °C

Ge 75 - 100 °C

iii. It forms its oxide SiO₂ very easily.

iv. Environmental friendly.

v. Si is highly stable. It does not change its behaviour while fabricating.

Si in elemental semiconductor stable.

Si \$500 B

GaAs \$15B

SiC, GaN \$5 B

LED \$15B

MEMS \$15B

* Si → indirect semiconductor

* GaAs → Direct B.G

↳ Does not emit light

↳ Comes into IR region

emits heat



- * SiC, GaN $\rightarrow$ for high power Applications
- * MEMS $\rightarrow$ micro electro mechanical system
- * ~~GaAs~~ GaAs $\rightarrow$ High speed
- * Mobility

$$\text{Si}(\mu_n) = 1500 \text{ cm}^2/\text{V-S}$$

$$\text{Ge}(\mu_n) = 4000 \text{ cm}^2/\text{V-S}$$

$$\text{High speed applicat?} \leftarrow \text{GaAs}(\mu_n) = \underline{8500} \text{ cm}^2/\text{V-S}$$

$$\boxed{\mu = v_d / E} ?$$

μ = mobility

v_d = Drift velocity

- * Mobility is the parameter for high speed applicat?

- * B.G $\rightarrow$ Energy gap b/w valence band and conduction band.

$$\text{B.G} \left\{ \begin{array}{l} \text{Si} = 1.12 \text{ eV} \\ \text{Ge} = 0.67 \text{ eV} \\ \text{GaAs} = 1.43 \text{ eV} \end{array} \right.$$

- * There are no free electrons in Si at 0°C.

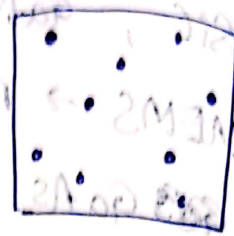
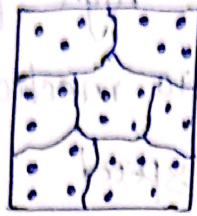
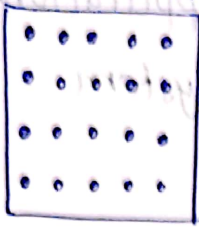
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- Si, GaAs $\rightarrow$ Diamond Shape?

- # VLSI fabricat? process

3 types of Si

- 1) Amorphous silicon $\rightarrow$ No crystal order, More defect
- 2) Poly Crystalline " $\rightarrow$ Multiple crystal order, less defect
- 3) Single Crystalline " $\rightarrow$ Uniform crystal order, Negligible defect



Single Crystal

Poly Crystalline

Amorphous

Crystal Growth and wafer preparatⁿ

- * Polycrystalline Si is used as starting material for the growth of device quality.
- * Single Crystal Si is used in the devices as it is the purest form.

Electronic Grade Silicon (EGS)

- ↳ Polycrystalline material of high quality.
- ↳ It is a raw material for the preparatⁿ of Single Crystal Si.
- ↳ It is 99.99% pure.
- ↳ EGS is obtained from quartzite (relatively pure form of sand SiO_2).

quartzite



(Metallurgical Grade Si) MGS

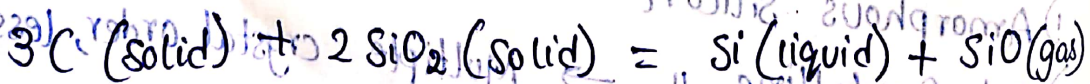


EGS

Steps

1) Obtaining EGS from Ore (Quartzite)

↳ Quartzite to MGS



* It is performed in submerged electrode in furnace filled with quartzite, coal, coke and wood chips.



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EGS to Single Crystal Silicon

step 1 : Czochralski Method (CZ)

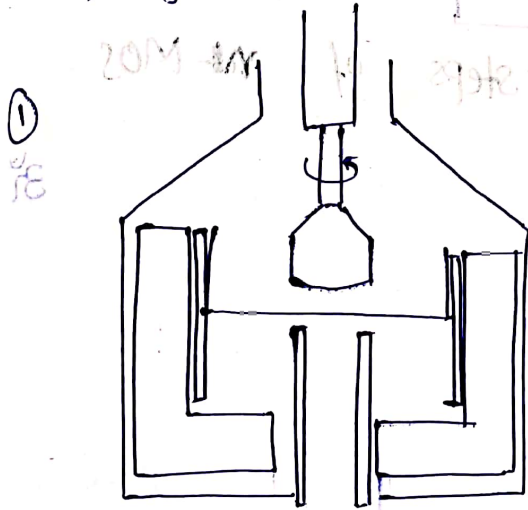
↳ Most used and used for both Si and GaAs

step 2 : Float zone Method (FZ)

↳ Very pure but high cost.

step 3 : Bridgeman technique

↳ for compound semiconductors



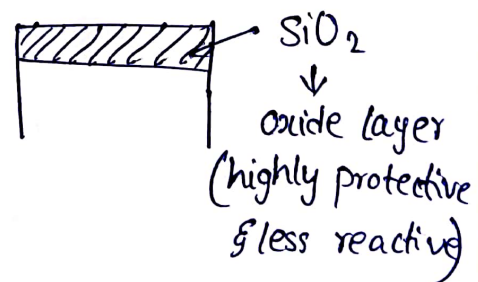
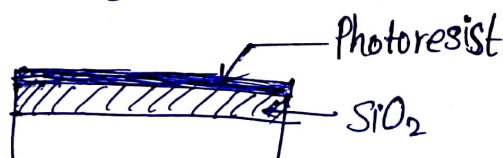
Diode fabrication :

① wafer preparation

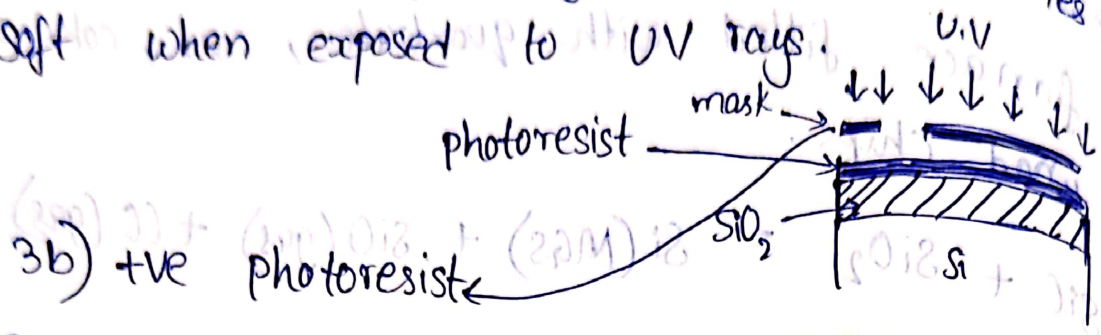
② oxidation

③ Lithography

④ 3a)

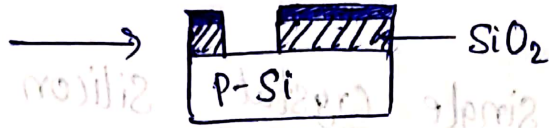


Photoresist is generally hard but becomes soft when exposed to UV rays.

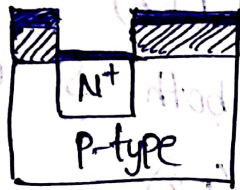


3b) +ve photoresist

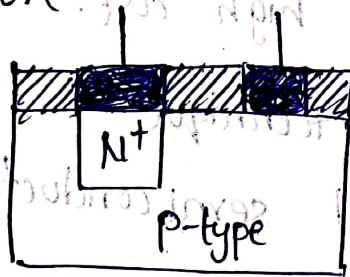
④ Etching



⑤ Diffusion

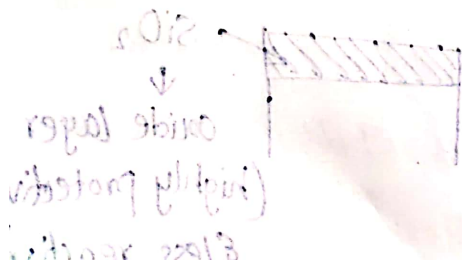
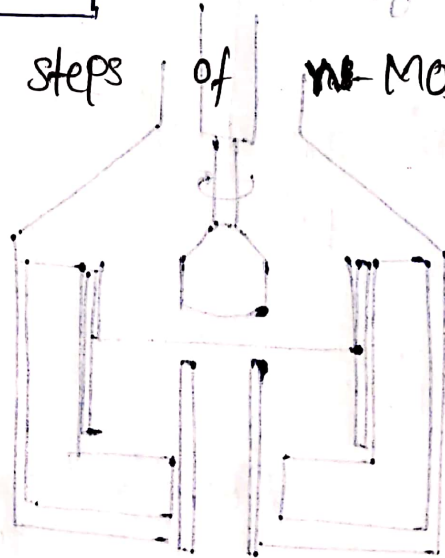


⑥ Metallization

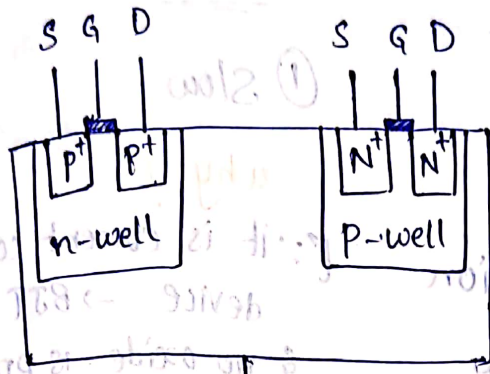


⑦ Packaging

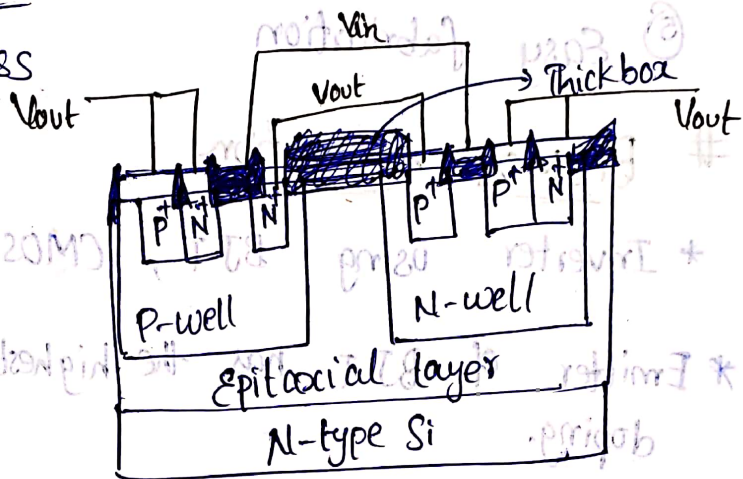
⑧ Write the fabrication steps of n-MOS.



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CMOS fabricatⁿ process



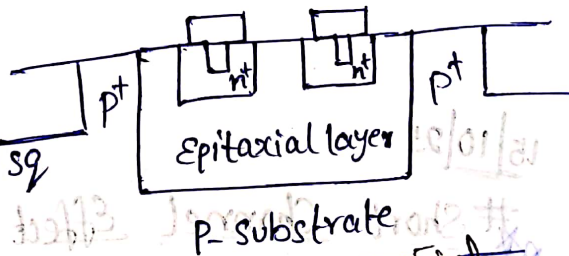
epitaxy
Twin tub Process



Resistor fabrication process

1) Epitaxial Resistor

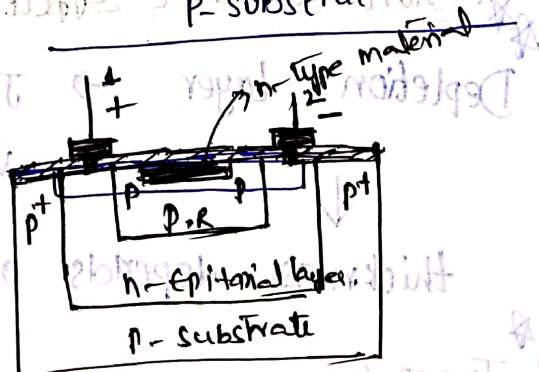
sheet resistance: $1 - 10 \text{ k}\Omega/\text{sq}$



2) Pinched Resistor (P.R.)

sheet resistance:

$10 - 50 \text{ k}\Omega/\text{sq}$



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MOSFET vs BJT

Why CMOS over MOS

Adv. of MOSFET

- ① Symmetrical
- ② Small in size
- ③ Less power consumption
- ④ High i/p impedance
- ⑤ Easy fabrication

Disadv.

① Slow

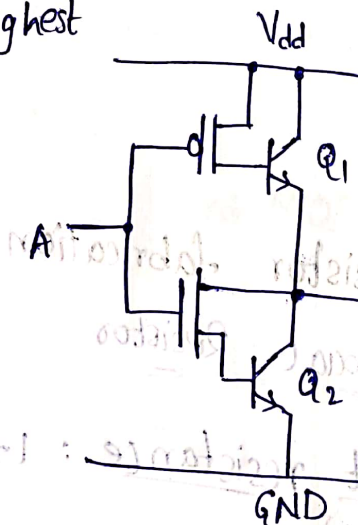
Why?

∵ it is current controlled device → BJT & no oxide is present to charge & discharge

BiCMOS : fabrication

* Inverter using BJT, CMOS and BiCMOS

* Emitter of BJT has the highest doping.



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Short Channel Effect

Depletion layer → J^n which contains $+ve$ & $-ve$ charged ions

thickness depends on doping conc.

* Types :

1. Sub-threshold current
2. Threshold voltage roll-off
3. Drain Induced Barrier Lowering (DIBL)

1. Punch Through
2. Velocity Saturation
3. Hot electron effect
4. Surface Scattering

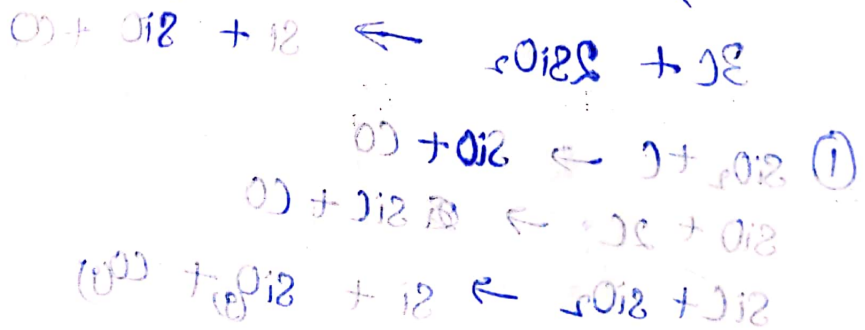


Amorphous $\rightarrow$ No crystal order (more defects)
 # Electronic grade Si $\rightarrow$ Polycrystalline with high purity
 * FET $\rightarrow$ Low material for prep. of single crystalline Si

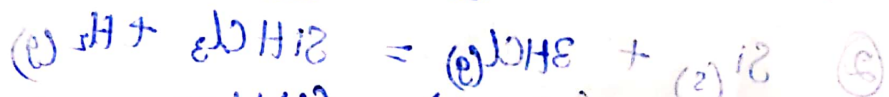
* FET is obtained from quartzite
 quartzite
 $\downarrow$
 MCD (metallurgical grade Si)
 $\downarrow$
 200

quartzite $\rightarrow$ Si $\rightarrow$ Silicon $\rightarrow$ water $\rightarrow$ water
 200 $\rightarrow$ 200 $\rightarrow$ 200

* Quartzite to water $\rightarrow$ performed in submerged electrodes and furnace (filled with quartzite and water coils)



$\downarrow$
 MCD $\rightarrow$ 200



SiHCl₃ (trichlorosilane) $\rightarrow$ liquid at room temp

Crystal growth and wafer prep. doping
 * Polycrystalline Si → single crystalline Si → Device
 multiple crystal order (less defects) uniform crystal order (Negligible defects)



• Amorphous → No crystal order (More defects)
 # Electronic Grade Si → EGS → Polycrystalline with high purity

* EGS → Raw material for prep. of single crystalline Si

↓
 99.9% pure

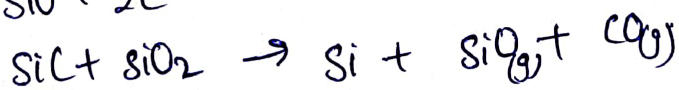
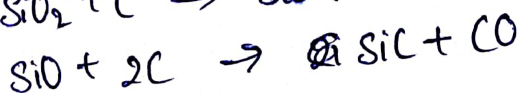
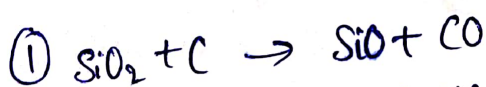
* EGS is obtained from quartzite

quartzite

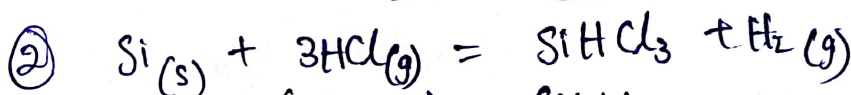
↓
 MGS (metallurgical grade Si)
 ↓
 EGS

quartzite Sand → Si → Silicon (eg Ingot) → Si wafer → wafer doping

• Quartzite to MGS → performed in submerged electrode arc furnace (filled with quartzite coal, coke and wood chips)



↓
MGS → 98% pure

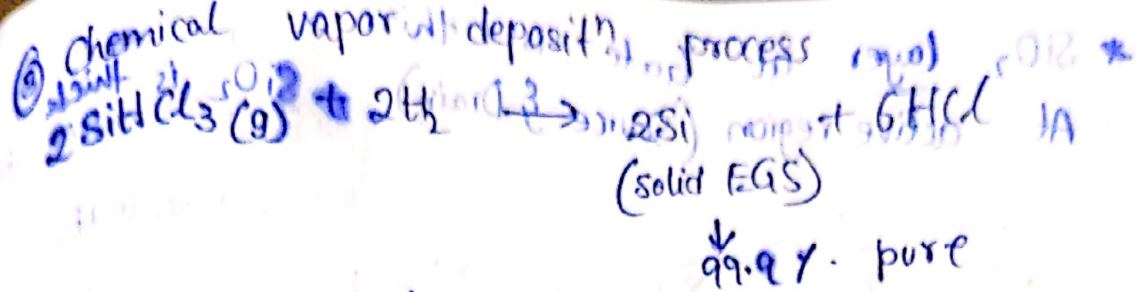


(anhydrous HCl)

(trichloro silane)

→ liquid at room temp





FGS $\rightarrow$ Single Crystalline Si

- 1) Czochralski method (CZ) $\rightarrow$ most used (for both Si & GaAs)
- 2) Float zone " (FZ) $\rightarrow$ very pure but high cost
- 3) Bridgman technique $\rightarrow$ Used for compound semiconductors (GaAs, GaN, etc)

Single Crystalline Si $\rightarrow$ mono crystalline Si.

form of Si where the entire solid has continuous, unbroken crystal lattice with no grain boundaries.

$\hookrightarrow$ highly pure $\hookrightarrow$ Its uniform structure property makes it ideal for high efficiency solar cells and semiconductor devices

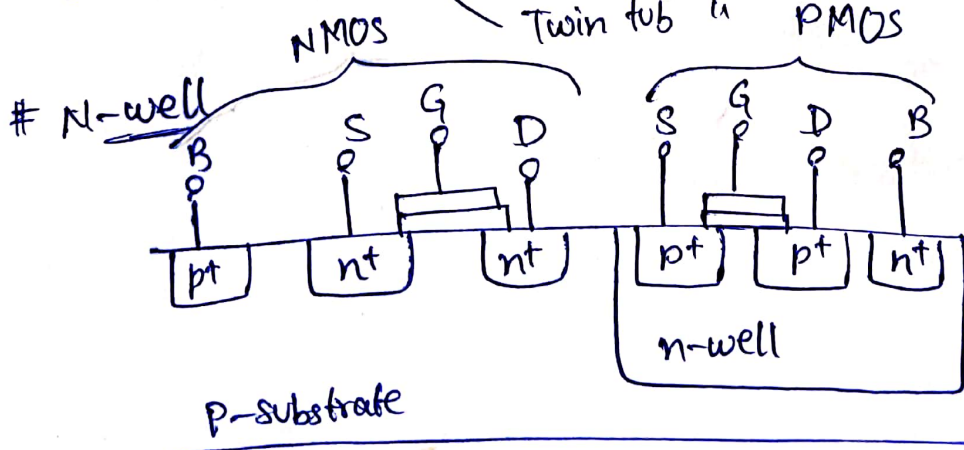
CZ

Si Cr. Si is produced by melting poly Cr. Si & then resolidifying it.

FGS is loaded into quartz crucible and heated at 1500°C along with (Ar) in presence of

CMOS fabricatⁿ

N-well process
P-well "
Twin tub "

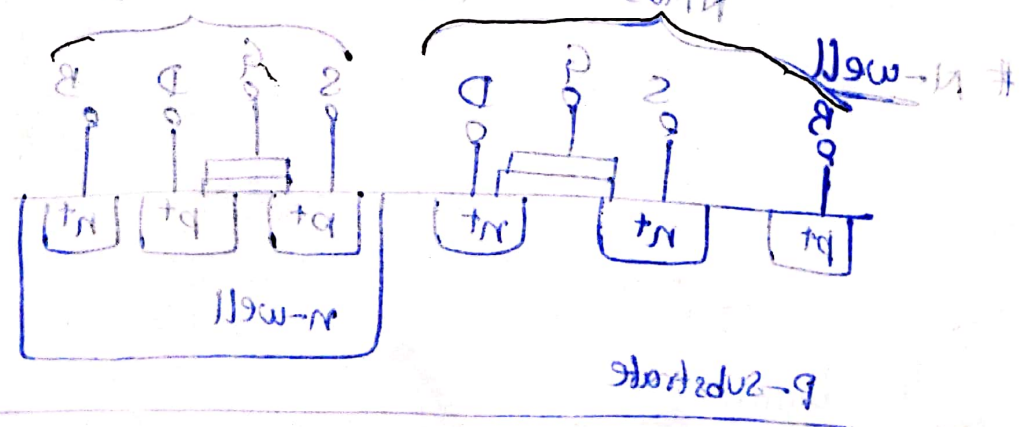


* SiO_2 layer at Gate is thinning
 At active region (Source & Drain) SiO_2 is thick

① (Clockwise method) (C) → single crystal
 (E) → most used (for both 2 & 3) SiO_2 layer
 (F) → need high cost
 (G) → used for compound semiconductor
 (H) → Bridge method
 (I) → Float gate
 (J) → Single crystal line 2 → mono crystalline 2
 (K) → Stick diagram
 (L) → Layout diagram
 (M) → Silicon

frontend
 backend
 broken crystal lattice no grain boundaries
 it is for high efficiency solar cells and semiconductor devices
 It is for high efficiency solar cells and semiconductor devices

Time domain → Algebraic domain
 Frequency
 * Effect of V_{th} in MOSFET
 If $V_{th} \uparrow$, $I_{D,sat} \downarrow$
 If $V_{th} \downarrow$, $I_{D,sat} \uparrow$
 to heat up the device and heat up the device



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VLSI Design rules :

- 1. Lambda (λ) based rules
- 2. Micron (μ) based rules

Layout Stick diagram

Wiring : The interconnect?

Local Global

speed
power
consumpt.

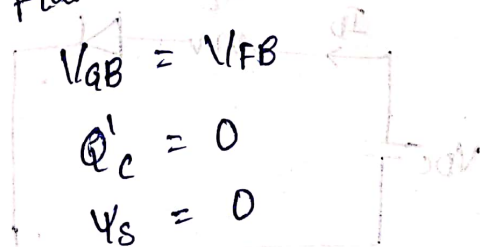
defects due to capacitance
in MOSFET (de to depletion layer)

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Basic SPICE level-1 MOSFET Model

* V_{th} CMOS

* Flat Band Condition

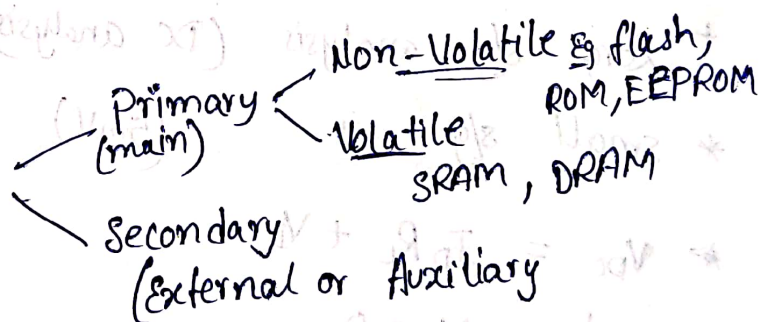


* Energy Band Diagram

Memory

* Data storage devices

* Types of memory



Photonics

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Displays

① Quantum Dot

② Carbon Nano tube & Graphene for flexible, transparent, and more efficient OLEDs and transistors.

③ Nanoparticles

①: → Improved color performance

③: → Improved conductivity
→ Durability

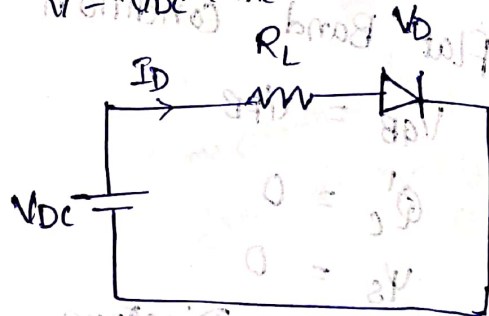
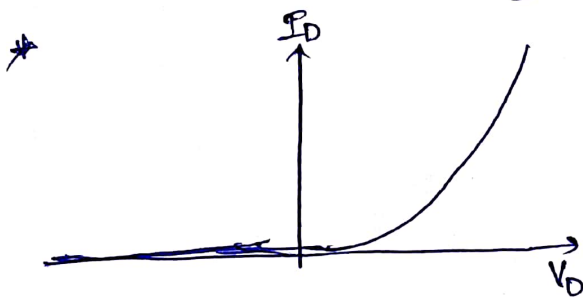
Wireless sensing and communication

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Derivation

$$I_D = I_0 \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$

$I_D \rightarrow$ Biasing current for DC
 $I = I_{DC} + I_{AC}$
 $V = V_{DC} + V_{AC}$



* Large s/g analysis (DC analysis) (5V)

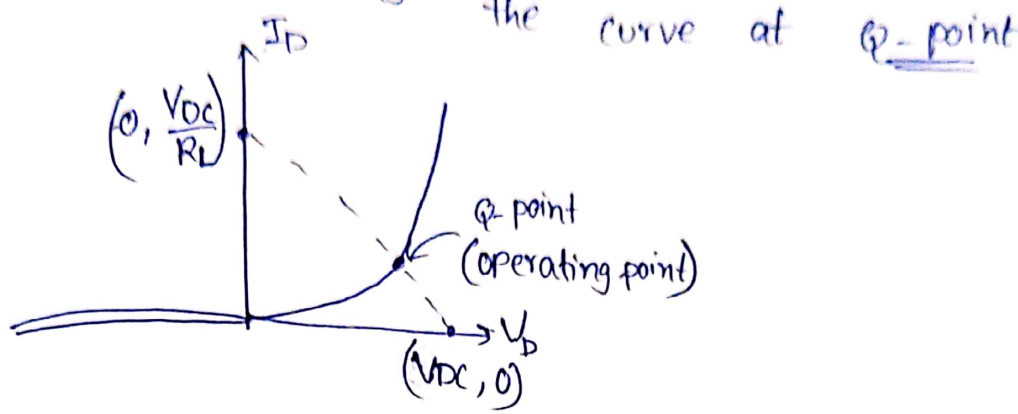
* small s/g analysis (5mV)

$$V_{DC} = I_D R_L + V_D$$

when $V_D = 0$

$$V_{DC} = I_D \cdot R_L$$

$$I_D = \frac{V_{DC}}{R_L}$$



$$* I_D = I_0 \left(e^{\frac{V_D}{\eta V_T}} - 1 \right) \quad I_D \rightarrow \text{Biasing current for DC}$$

$$I = I_{DC} + I_{AC}$$

$$V = V_{DC} + V_{AC}$$

$$I_D = I_0 \left(e^{\frac{V_{DC} + V_{AC}}{\eta V_T}} - 1 \right) = I_0 \left(\underbrace{e^{\frac{V_{DC}}{\eta V_T}}}_{\text{large } 5V \rightarrow} \cdot \underbrace{e^{\frac{V_{AC}}{\eta V_T}}}_{\text{small } 5mV \rightarrow} - 1 \right)$$

$\gg 1$

~~$$I_D = I_0 \left(e^{\frac{V_{DC}}{\eta V_T}} - 1 \right)$$~~

$$I_D = I_0 \left(e^{\frac{V_{DC}}{\eta V_T}} \cdot e^{\frac{V_{AC}}{\eta V_T}} \right)$$

$$= I_0 \cdot e^{\frac{V_{DC}}{\eta V_T}} \cdot \left(1 + \frac{V_{AC}}{\eta V_T} \right) \left(\because e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right)$$

$x \text{ is small} \Rightarrow e^x = 1 + x$

$$I_D = I_0 e^{\frac{V_{DC}}{\eta V_T}} + I_0 e^{\frac{V_{DC}}{\eta V_T}} \cdot \frac{V_{AC}}{\eta V_T}$$

$$= I_{DC} + I_{DC} \cdot \frac{V_{AC}}{\eta V_T}$$

$$I_D = I_{DC} + I_{AC}$$

$$I_{AC} = I_{DC} \cdot \frac{V_{AC}}{\eta V_T}$$

$$\frac{V_{AC}}{I_{AC}} = \frac{I_{DC}}{\eta V_T}$$

$$\boxed{r_{AC} = \frac{I_{DC}}{\eta V_T} = r_d} \rightarrow \text{dynamic resistance}$$

static resistance?